

Nebulised therapies for smoke inhalation injury: mechanistic rationale, evidence gaps, and the case for an *adaptive platform trial*.

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01 · SCALE OF THE PROBLEM

Smoke inhalation injury is the *single largest preventable driver* of burn mortality.

35%

BURN ADMISSIONS WITH INHALATION INJURY¹

38%

SUBSEQUENT PNEUMONIA RATE¹

+20%

ABSOLUTE MORTALITY INCREASE¹

ZERO

THERAPIES OF PROVEN BENEFIT²

03 · TWO-HIT PATHOPHYSIOLOGY

A distinct *two-hit* injury cascade — a mechanistic rationale for each therapy.

FIRST HIT · 0–6 HOURS

Chemical ciliotoxicity

INTACT

DENUDED

► Direct chemical injury to epithelium

► Ciliary destruction + denudation

► Loss of mucociliary clearance

► Neutrophil influx + cytokine storm

→ Ref 2

SECOND HIT · 12–72 HOURS

Fibrin-cast obstruction

FIBRIN-RICH AIRWAY CAST

► Bronchial blood flow ↑ 10-fold

► Plasma exudation → coagulopathy

► Fibrin + cellular debris casts

► Small-airway obstruction · V/Q mismatch

→ Ref 2

NEBULISED THERAPEUTIC TARGETS

Three mechanistic anchors

Nebulised unfractionated heparin

Attenuates second-hit fibrin coagulopathy^{2,3}

N-Acetylcysteine

Mucolytic + antioxidant; pairs with heparin³

β-agonist (albuterol)

Bronchodilation + mucociliary clearance

04 · AIM

Why this synthesis exists.

To synthesise contemporary preclinical, translational, and clinical evidence for nebulised **heparin**, **NAC**, and **β-agonists** in smoke inhalation injury.

- 01 **Mechanistic plausibility** — does the biology hold up?
- 02 **Methodological limits** — why have RCTs failed?
- 03 **Trial design** — what methodology could finally answer?

05 · METHOD

Narrative synthesis approach.

Search. PubMed, Embase, Cochrane (inception–April 2026); inhalation/smoke injury × each candidate therapy.

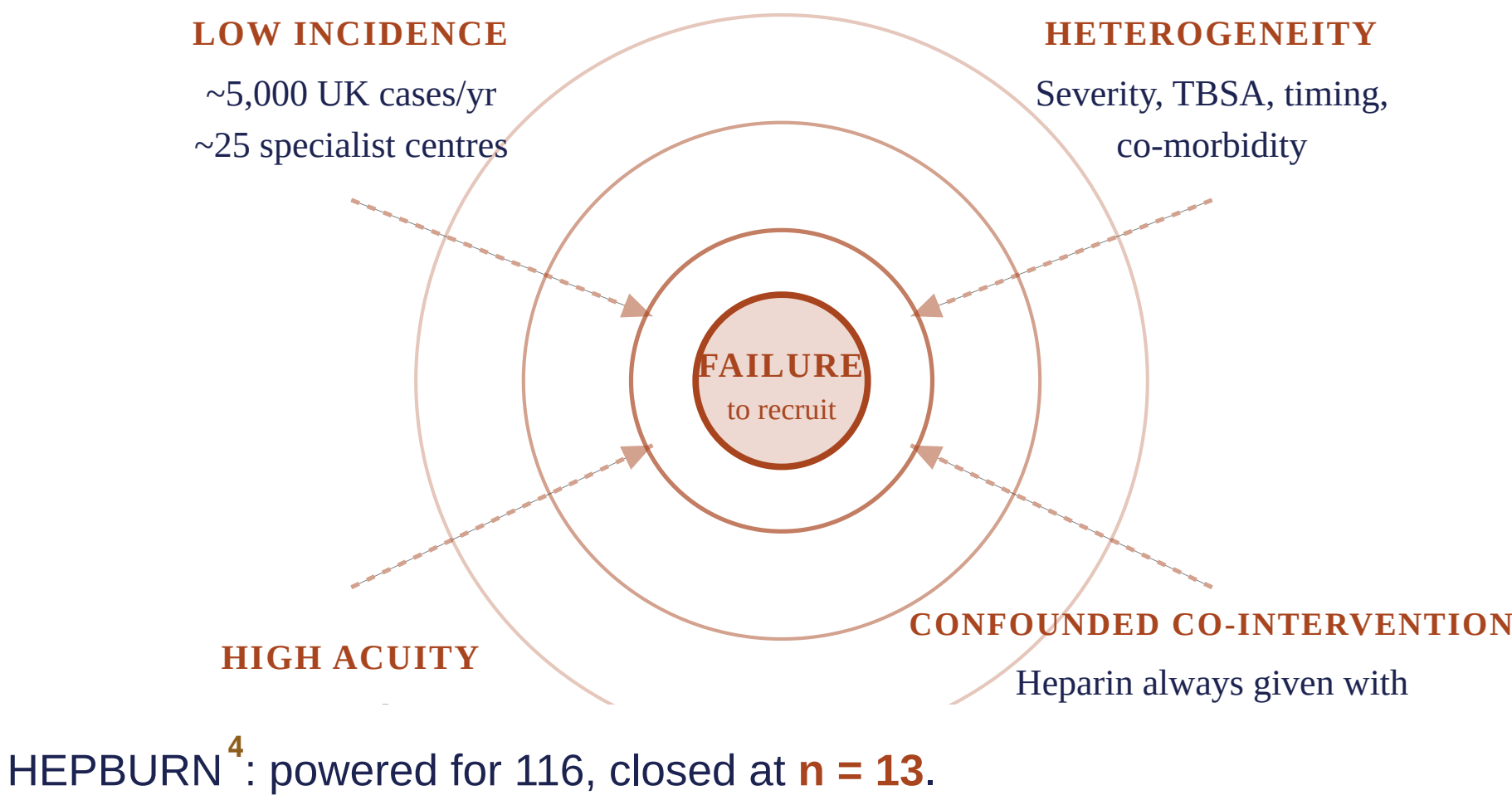
Inclusion. Observational cohorts, RCTs, and existing SR/meta-analyses in adult burn populations.

Appraisal. Studies appraised narratively — no quantitative pooling (heterogeneity precludes it).

Synthesis. Findings grouped by intervention and trial-design limitation.

06 · WHY RCTS FAIL

Four structural barriers — together, fatal.



07 · THE CLINICAL EVIDENCE BASE

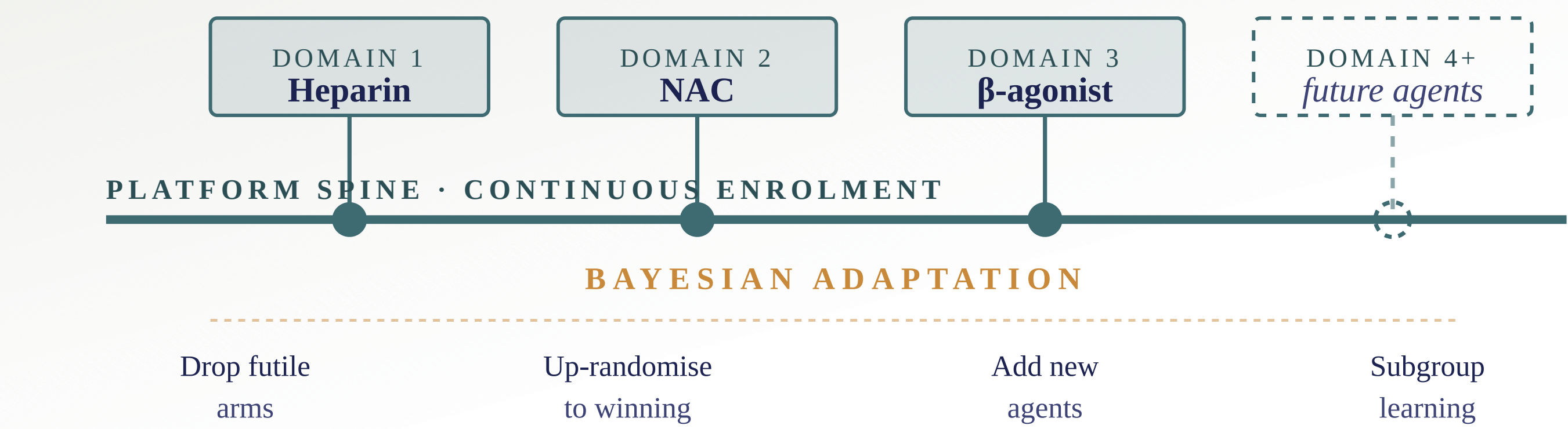
A 27-year evidence base — *and a flagship RCT that collapsed under its own weight*.

STUDY (YEAR)	FINDING	STATUS
Desai 1998	Heparin + NAC ↓ atelectasis, reintubation, mortality (paediatric retrospective).	↓ MORTALITY
Enkhbaatar 2007	Aerosolised anticoagulants ↓ acute lung injury (ovine burn+smoke).	PRECLIN. SIGNAL
Miller 2009	Heparin/NAC ↑ LIS, oxygenation, survival (n=30 adults, retrospective).	↓ MORTALITY
Elsharnouby 2014	10,000 IU vs 5,000 IU heparin: higher dose ↓ LIS & vent duration (n=29).	DOSE SIGNAL
Glas 2014–2020 · HEPBURN	International RCT. Powered 116. Stopped at n=13 .	TERMINATED

Twenty-seven years of investigation. Preclinical signal: strong. Clinical evidence: heterogeneous, single-centre, retrospective, and — when scaled up — *unable to recruit*.

08 · THE ADAPTIVE PLATFORM SOLUTION

A single, perpetual platform — answering many questions at once.



Master protocol, multi-centre. Patients consented once into the platform; randomised across domains. Bayesian updating drops futile arms early. REMAP-CAP proof of concept: 13,000+ patients enrolled globally during COVID-19.⁶

PROPOSAL

A UK-led adaptive platform trial for nebulised therapies in smoke inhalation injury — embedded across the BBA-affiliated specialist burn centre network.

PLATFORM DESIGN

Master protocol; three initial domains (heparin, NAC, β-agonist); Bayesian adaptive randomisation; pre-specified futility / efficacy stopping rules.⁶

PRIMARY OUTCOME

28-day ventilator-free days (compatible with HEPBURN⁴ & Hakim 2025⁵); pneumonia, mortality, ICU LoS as secondary.

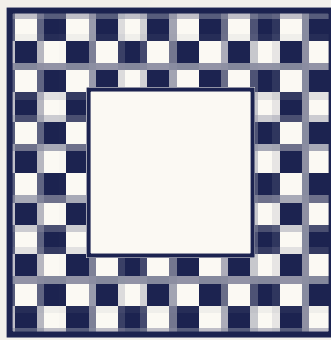
FEASIBILITY ADVANTAGE

One consent, one infrastructure, one statistical core. Smaller per-domain samples; perpetual recruitment overcomes the HEPBURN failure mode.

FULL BIBLIOGRAPHY & DOIS

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All 20 references verified · maintained alongside the live poster microsite



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